Wearable Device Chipset Identification and Security Assessment Report

Compiled research notes

24 October 2026

Abstract

This document consolidates hardware identification and public vulnerability research for selected wearable and heart-rate monitoring devices. Primary sources include direct PCB photography, FCC filings, manufacturer documentation, and published security research. Two Garmin identifiers are treated as distinct per user correction. A generic unbranded smartwatch is also included from packaging evidence.

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1 Device and Chipset Summary

Device	Model / Identifier	Primary SoC / Chipset	Notes / Source
Fitbit Charge 6 (Porcelain)	G3MP5 / GA05183 / FB422 family	Ambiq Micro Apollo 4 Blue Plus (AMA4B2KP-KXR)	Ultra-low-power SoC, BLE 5.0, GPS+GLONASS. TechInsights teardown.
Garmin 03711-00 VER 2	03711-00 VER 2 (A03711, FCC ID IPH-03711)	Nordic Semiconductor nRF52832-QFAA (E0 revision). Marking: N52832 QFAAE0 2448JF	Confirmed from user-supplied PCB photograph and silkscreen. Multiprotocol BLE + ANT+ capable.
Garmin HRM-Dual	010-12883-00	Not confirmed in this report (treated as distinct from 03711-00 VER 2)	Official retail part number. Dual ANT+ + Bluetooth LE transmission.
Polar H10	Standard H10 sensor	Nordic Semiconductor nRF52832 (or closely related nRF52 series)	Dual BLE + ANT+ + 5 kHz GymLink, ECG + accelerometer. Polar documentation and board analysis.
Generic / Unbranded Smartwatch	Packaging ID: 5H0305 ("Stop Seller No. 5H0305")	Unknown (typical budget Chinese BLE SoC — often Realtek, JL/JieLi, or similar low-cost multiprotocol MCU)	Square face, black strap, blue geometric watch face. Generic "SMARTWATCH" packaging. Common private-label fitness/Bluetooth-call model.

2 Garmin Model Distinction

The identifiers **03711-00 VER 2** and **HRM-Dual (010-12883-00)** are treated as separate entries following user correction.

- The PCB photograph clearly shows silkscreen `GARMIN 03711-00 VER 2` together with the Nordic nRF52832 (QFAAE0 marking).
- The commercial HRM-Dual product is sold under part number 010-12883-00.
- The 03711 designation appears in the 2019 FCC filing IPH-03711 for a "Wireless Heart Rate Monitor," but the two are kept distinct in this report.

3 Security and CVE Analysis

3.1 Nordic nRF52832 (Garmin 03711-00 VER 2 and Polar H10)

- **2025 research:** Remote RF side-channel attack recovers the full 128-bit AES key from RF emissions at approximately 1 m distance. Requires on the order of 90 000 traces and machine-learning analysis. Practical but time-intensive (typically days of continuous transmission). No formal CVE number widely assigned at the time of writing, but the vulnerability is documented in the security literature.
- The nRF52832 lacks ARM TrustZone / CryptoCell (present on the nRF52840). Secure-boot support is limited; the chip can be fully erased/recovered via SWD.

3.2 Ambiq Apollo 4 Blue Plus (Fitbit Charge 6)

Affected by Cordio Bluetooth Low Energy Host stack issues shared across multiple Ambiq Apollo families:

- CVE-2024-48981
- CVE-2024-48982
- CVE-2024-48983
- CVE-2024-48984
- CVE-2024-48985
- CVE-2024-48986

Ambiq has released "Known Defects" patches via its content portal. Some issues were addressed as early as 2021; others in early 2026 releases. Impact depends on the specific firmware revision used by Fitbit.

3.3 Garmin HRM-Dual (010-12883-00)

No device-specific public CVEs identified beyond any underlying silicon issues (chipset not confirmed in this report).

3.4 Polar H10

No device-specific public CVEs located. Polar documentation notes privacy considerations when dual Bluetooth / ANT+ broadcasting is enabled (any compatible receiver can obtain the signal). Later firmware versions (e.g., 3.2.0, 4.0.4) added connectivity fixes and secure-pairing features. Polar publishes an official BLE SDK.

3.5 Generic / Unbranded Smartwatch (5H0305)

Exact chipset is not identifiable from packaging alone, therefore specific silicon CVEs cannot be assigned. Devices in this class commonly exhibit:

- Weak or missing Bluetooth pairing / encryption.
- Insecure companion applications (hard-coded credentials, lack of certificate pinning, excessive permissions).
- Absence of firmware signing or secure boot, facilitating reflashing or reverse engineering.

No formal CVEs are published against the specific "5H0305" identifier because these units are unbranded and change suppliers frequently.

4 Risk Summary

Risk Level	Issue	Affe
Medium	RF side-channel AES key recovery (proximity + significant data collection required)	Garr
Low–Medium	Cordio BLE stack vulnerabilities (patch status firmware-dependent)	Fitbi
Low	Privacy leakage via open broadcast modes	Pola
Elevated (class)	Weak pairing, insecure apps, no secure boot	Gene

5 Recommendations

1. Keep firmware updated on all name-brand devices via the respective manufacturer applications (Polar Flow/Beat, Fitbit/Google, Garmin Connect where applicable).
2. For the Polar H10, disable dual Bluetooth or open broadcasting when not required; perform a factory reset before disposal or transfer.
3. Physical access / SWD recovery remains feasible on nRF52832-based devices.
4. Treat the unbranded 5H0305 unit as higher risk until the actual SoC and firmware version can be identified (FCC ID on the device, BLE advertisement data, or teardown).

6 Sources and Notes

- Direct PCB photography supplied by the user (Garmin 03711-00 VER 2 and Polar H10 boards).
- FCC ID IPH-03711 filing and associated exhibits.
- Official Garmin product listings for part number 010-12883-00.
- TechInsights teardown data for Fitbit Charge 6.
- Nordic Semiconductor product documentation and 2025 side-channel research literature.
- Ambiq security advisories covering Cordio stack CVEs.
- Polar H10 user documentation and firmware release notes.

This report reflects public information available as of 24 October 2026. Chipset identification for the generic smartwatch remains provisional pending further physical examination.